AUTOMATED PNEUMONIA DETECTION FROM CHEST X-RAY IMAGES USING CONVOLUTIONAL NEURAL NETWORKS

Govind Ramesh, Kamran Khan, & Aria Sharma

1 INTRODUCTION

Pneumonia is a leading cause of death globally, particularly in children and elderly populations, making early and accurate diagnosis critical for effective treatment. Chest X-rays are the primary diagnostic tool used by radiologists to identify pneumonia, but interpreting these images requires significant expertise and is time consuming. In resource-limited settings or areas without access to a radiologist, delays in diagnosis can lead to worse patient outcomes. Deep learning models have shown strong promise in assisting with medical image analysis by detecting subtle patterns that indicate disease presence, potentially serving as a diagnostic aid to reduce missed cases and help prioritize patient care.

In this project, we develop and analyze deep learning models for automated pneumonia detection from chest X-ray images. Each model takes a chest X-ray image as input, resized to 224×224 and converted to three channels, and outputs a binary label indicating pneumonia presence or absence along with a probability score between 0 and 1. Models are trained using ground-truth labels from the Kaggle Chest X-ray Pneumonia dataset under a supervised learning framework. We experiment with CNN, ResNet, and Vision Transformer architectures, and use weighted binary cross-entropy as the primary loss function to handle class imbalance. Our emphasis is on model interpretability, handling class imbalance, and evaluating clinical reliability through metrics like sensitivity and specificity that matter for real-world deployment.

2 RELATED WORK

Several important works have laid the foundation for using deep learning in pneumonia detection from chest X-rays. Wang et al. (2017) released the ChestX-ray14 dataset, providing over 112,000 chest X-ray images labeled across 14 thoracic diseases including pneumonia, which became a standard benchmark for medical image classification. Building on this, Rajpurkar et al. (2017) developed CheXNet, a 121-layer DenseNet that achieved radiologist-level performance on pneumonia detection with an F1 score of 0.435. Their model also used Class Activation Maps to visualize which X-ray regions influenced predictions, demonstrating that deep learning could match expert performance. However, CheXNet focused primarily on accuracy without deeply analyzing model behavior or addressing challenges like class imbalance.

Key architectures referenced in prior work include DenseNet (Huang et al., 2017) and ResNet (He et al., 2016). DenseNet connects each layer to all previous layers, improving gradient flow and feature reuse, and was used in CheXNet. ResNet uses skip connections to enable effective training of very deep networks, and we adopt ResNet-50 as our primary transfer learning baseline. For interpretability, we use Grad-CAM (Selvaraju et al., 2017), which highlights the image regions most important for a given prediction. Our work differs from prior approaches by placing equal emphasis on understanding model behavior alongside achieving strong performance, providing deeper analysis of feature transfer, class imbalance handling, and clinical readiness.

We conduct our experiments on the Kaggle Chest X-ray Pneumonia dataset Mooney (2018), which contains 5,863 labeled JPEG chest X-rays split into *PNEUMONIA* and *NORMAL* categories across provided train, validation, and test folders. The dataset exhibits class imbalance, with pneumonia cases outnumbering normal cases roughly 3:1 in the training split. Our original proposal planned to use the ChestX-ray14 dataset filtered for Pneumonia vs. No Finding with a custom 80/10/10 split, but we switched to the Kaggle dataset because it provides cleaner pre-defined splits, a simpler binary label structure, and a more manageable scale for our timeline.

3 Problem Decomposition & Technical Approach

Our pipeline covers three goals: a reproducible data pipeline, a progression of models from simple to sophisticated, and rigorous interpretability analysis.

3.1 Data Pipeline

We use the Kaggle Chest X-ray Pneumonia dataset, organized into `train/val/test` splits with `NORMAL` and `PNEUMONIA` subdirectories. This replaces the ChestX-ray14 dataset originally proposed, as the Kaggle dataset offers pre-defined patient splits and a straightforward binary label structure better suited to our midterm scope. Our `prepare_data.py` script constructs a unified manifest CSV preserving the provided splits to maintain patient-level separation and avoid data leakage. Preprocessing is standardized: images are resized to 224×224, grayscale images converted to three channels, and pixel values normalized. Training-time augmentation is applied only to the training split.

3.2 Model Hierarchy

We evaluate three architectures in increasing order of sophistication:

Baseline 1: Small CNN. Trained from scratch using weighted BCE to handle class imbalance. Serves as the lower-bound reference for transfer learning benefits.

Baseline 2: ResNet-50. Initialized from ImageNet weights and fine-tuned end-to-end. Tests whether natural image features transfer to medical radiographs. Evaluated under both standard and weighted BCE to test Hypothesis H2.

Proposed Method: DINOv2 Linear Probe. A frozen DINOv2 ViT-S/14 encoder with a single linear classification head. DINOv2 is a self-supervised Vision Transformer producing rich patch-level features with strong cross-domain transferability. Keeping the encoder frozen isolates representation quality from fine-tuning dynamics. Evaluated under both BCE and weighted BCE.

3.3 Hypotheses

- **H1:** A frozen DINOv2 linear probe will achieve higher ROC-AUC and F1 than fine-tuned ResNet-50 on the same split.
- **H2:** Weighted BCE will improve pneumonia-class recall and PR-AUC over standard BCE under the same architecture.
- **H3:** DINOv2 explanation maps will show stronger faithfulness than Grad-CAM, measured by deletion/insertion AUC on correctly classified test images.

3.4 Interpretability

For CNN and ResNet-50, we apply Grad-CAM to the final convolutional layer to produce spatial heatmaps. For DINOv2, we generate patch-feature similarity maps and attention rollout visualizations. We also compute faithfulness metrics (deletion and insertion curves) to quantitatively evaluate whether highlighted regions are causally important for predictions. Side-by-side Grad-CAM vs. DINO comparisons will be included in the final report.

3.5 Evaluation Procedure

All models are evaluated on the same held-out test split using a consistent protocol. The dataset's provided train/validation/test folders are preserved without modification to avoid leakage. The classification threshold is selected by optimizing F1 on the validation set rather than using a fixed 0.5 cutoff, which is important under class imbalance. Test set performance is reported using accuracy, ROC-AUC, PR-AUC, F1, precision, recall (sensitivity), specificity, and confusion matrix counts (TP, TN, FP, FN). PR-AUC is treated as the primary metric given the class imbalance, as it is more informative than ROC-AUC in imbalanced settings.

3.6 Current Status

Due to computational constraints, only the CNN with weighted BCE has been completed so far, with the remaining experiments actively in progress. The full experiment matrix is:

- CNN + weighted BCE **(completed)**
- ResNet-50 + BCE *(in progress)*
- ResNet-50 + weighted BCE *(in progress)*
- DINOv2 linear + BCE *(in progress)*
- DINOv2 linear + weighted BCE *(in progress)*

4 Experiments & Results

4.1 CNN Baseline Results

We train a small CNN from scratch with weighted BCE as our first baseline. A from-scratch CNN is appropriate here because it establishes the minimum performance achievable without any pre-trained features or external data, giving a clear lower bound against which transfer learning and self-supervised methods can be measured. Table 1 summarizes test set results at a validation-selected threshold of 0.4107.

Metric	Value
Accuracy	0.8253
Precision	0.8980
Recall (Sensitivity)	0.8128
Specificity	0.8462
F1 Score	0.8533
ROC-AUC	0.8955
PR-AUC	0.9242

Table 1: CNN baseline (`cnn_weighted_bce`) test set performance, best validation epoch 7.

The confusion matrix shows TP = 317, TN = 198, FP = 36, FN = 73. High precision (0.8980) indicates reliable positive predictions, while recall (0.8128) reveals that roughly 18% of pneumonia cases are missed, which is a clinically relevant gap since false negatives carry higher risk in a screening setting. ROC-AUC of 0.8955 and PR-AUC of 0.9242 indicate strong class separability, with PR-AUC especially informative given class imbalance. Figure 1 shows the ROC curve, Precision-Recall curve, and confusion matrix.

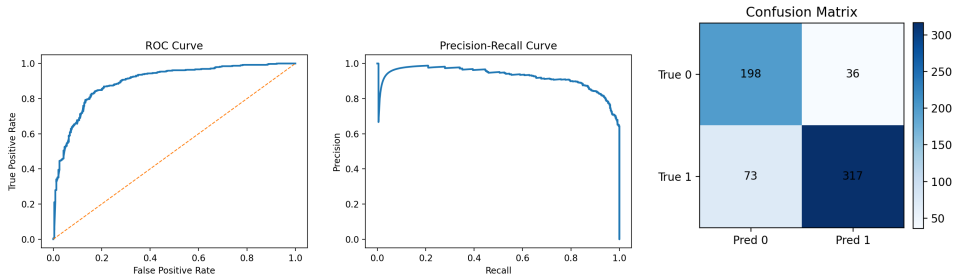

Figure 1: ROC curve, PR curve, and confusion matrix for the CNN baseline.

4.2 Grad-CAM Interpretability

We apply Grad-CAM to visualize which regions influenced each prediction. Figure 2 shows three representative test examples. The first (IM-0001-0001) shows strong, well-localized activation

across the lung fields, consistent with clinically relevant consolidation patterns. The second and third examples (IM-0003-0001, IM-0005-0001) show weaker, edge-concentrated saliency maps suggesting the model sometimes relies on boundary cues rather than lung parenchyma. This motivates our planned comparison with DINOv2 attention maps, which are expected to produce more semantically grounded localizations.

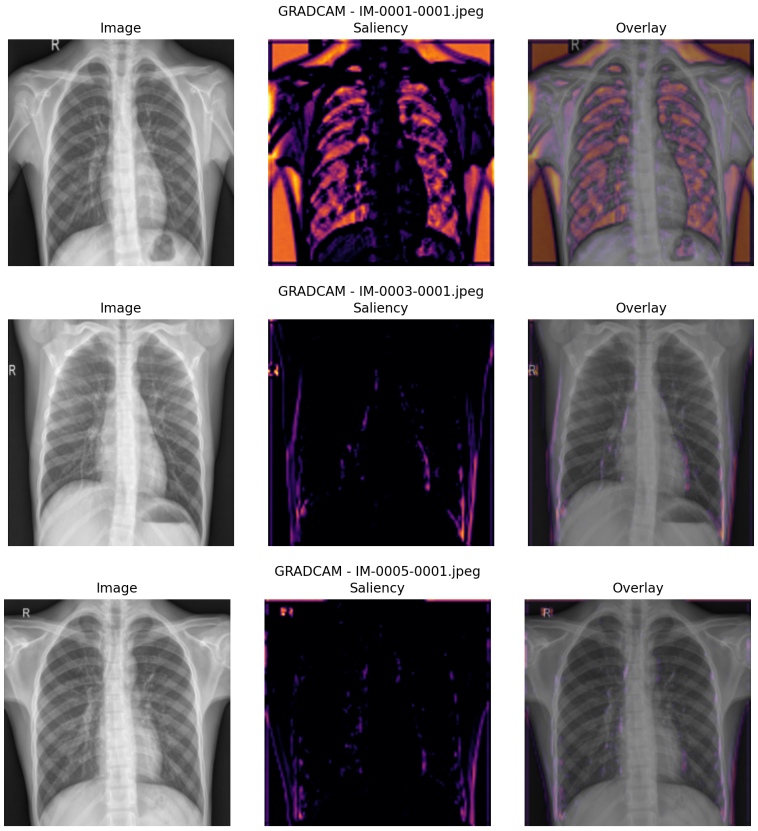

Figure 2: Grad-CAM examples: original X-ray, saliency map, and overlay. Top row shows strong lung-field activation; bottom rows show weaker, less localized attention.

4.3 Ongoing Experiments and Interpretability Extensions

ResNet-50 and DINOv2 experiments are in progress and results will be reported in the final submission. Upon completion, we will produce a unified metrics table comparing all five configurations to evaluate H1 and H2. For interpretability, we will generate side-by-side Grad-CAM vs. DINO explanation maps and report deletion/insertion AUC as a faithfulness score to evaluate H3. In terms of method progress, the full DINOv2 pipeline including the frozen encoder, linear head, training loop, and interpretability scripts has been implemented and verified on a small smoke-test config, so the model is ready to run once compute is available.

5 Next Steps

The remaining work breaks down into three phases. In the first two weeks, we will complete the ResNet-50 runs (BCE and weighted BCE) and begin ResNet-50 interpretability analysis. In the following two weeks, we will run the DINOv2 experiments and generate all explanation maps and faithfulness curves. In the final week, we will consolidate results into a unified comparison table, complete error analysis using `top_correct.csv` and `top_errors.csv`, and finalize all report visuals.

The key potential challenge is compute availability, since ResNet-50 and DINOv2 are both significantly slower to train than the CNN baseline. As a mitigation, we have prioritized the weighted BCE variants since they are more directly tied to our hypotheses. A secondary challenge is that DINOv2 attention maps may require additional tuning on grayscale medical images, since the model was pretrained on natural RGB images.

Final success will be evaluated by: (1) whether DINOv2 achieves higher ROC-AUC and F1 than ResNet-50 (H1); (2) whether weighted BCE improves recall and PR-AUC over standard BCE (H2); and (3) whether DINOv2 faithfulness scores exceed Grad-CAM scores on the deletion/insertion curves (H3).

REFERENCES

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.

Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4700–4708, 2017.

Rohit Kundu, Repayan Das, Zong Woo Geem, Gi-Tae Han, and Ram Sarkar. Pneumonia detection in chest X-ray images using an ensemble of deep learning models. *PLOS ONE*, 16(9):e0256630, 2021. doi: 10.1371/journal.pone.0256630.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)*, pp. 2980–2988, 2017.

Paul Mooney. Chest X-ray images (pneumonia), 2018. URL `https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia`.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2023.

Pranav Rajpurkar, Jeremy Irvin, Kaylie Zhu, Brandon Yang, Hershel Mehta, Tony Duan, Daisy Ding, Aarti Bagul, Curtis Langlotz, Katie Shpanskaya, Matthew P. Lungren, and Andrew Y. Ng. CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv preprint arXiv:1711.05225*, 2017.

Wojciech Samek, Alexander Binder, Grégoire Montavon, Sebastian Lapuschkin, and Klaus-Robert Müller. Evaluating the visualization of what a deep neural network has learned. *IEEE Transactions on Neural Networks and Learning Systems*, 28(11):2660–2673, 2017.

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626, 2017.

Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2097–2106, 2017.